

[EOD REPORT] Adhitya Charan - 15-04-2026

1 message

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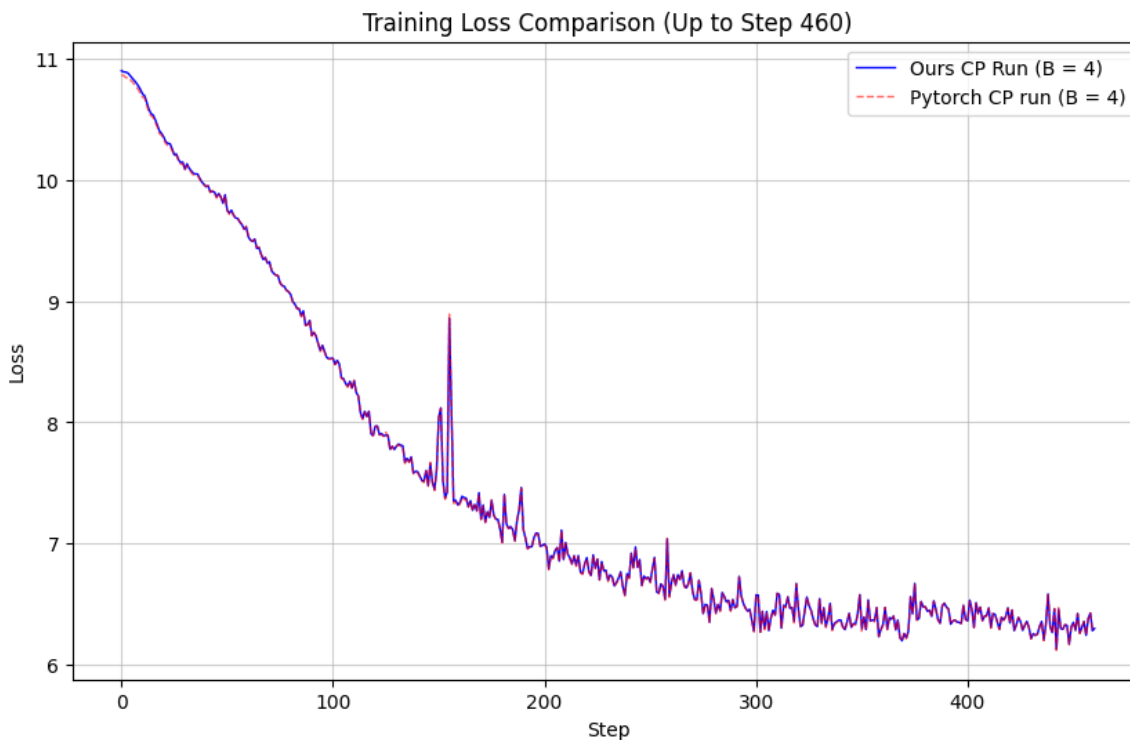
15 April 2026 at 21:11

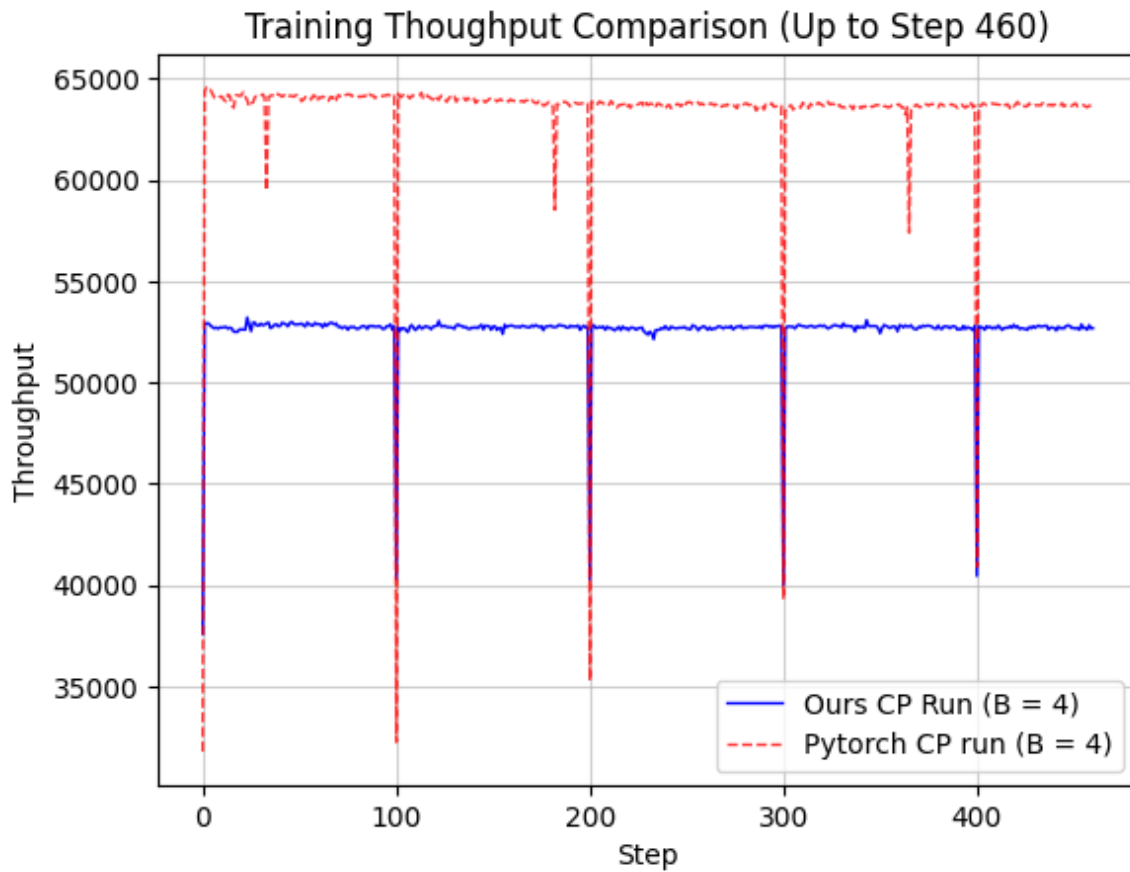
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Subject : Partial Updates of gradients in SPDA backward pass

Today I tried to look into the sdpa merger to see if the error was there. But there was no error there as well and the loss was still exploding with the norm ,
so I decided to revert to the last push to my git branch and build from that since that was working with 52k throughput without loadbalancing at least.





I looked into SDPA backward implementation in pytorch's implementation and found that there were two areas where `partial_Update` is called not just 1. Partial updation is used forward and backward differently

In Forward Pass:

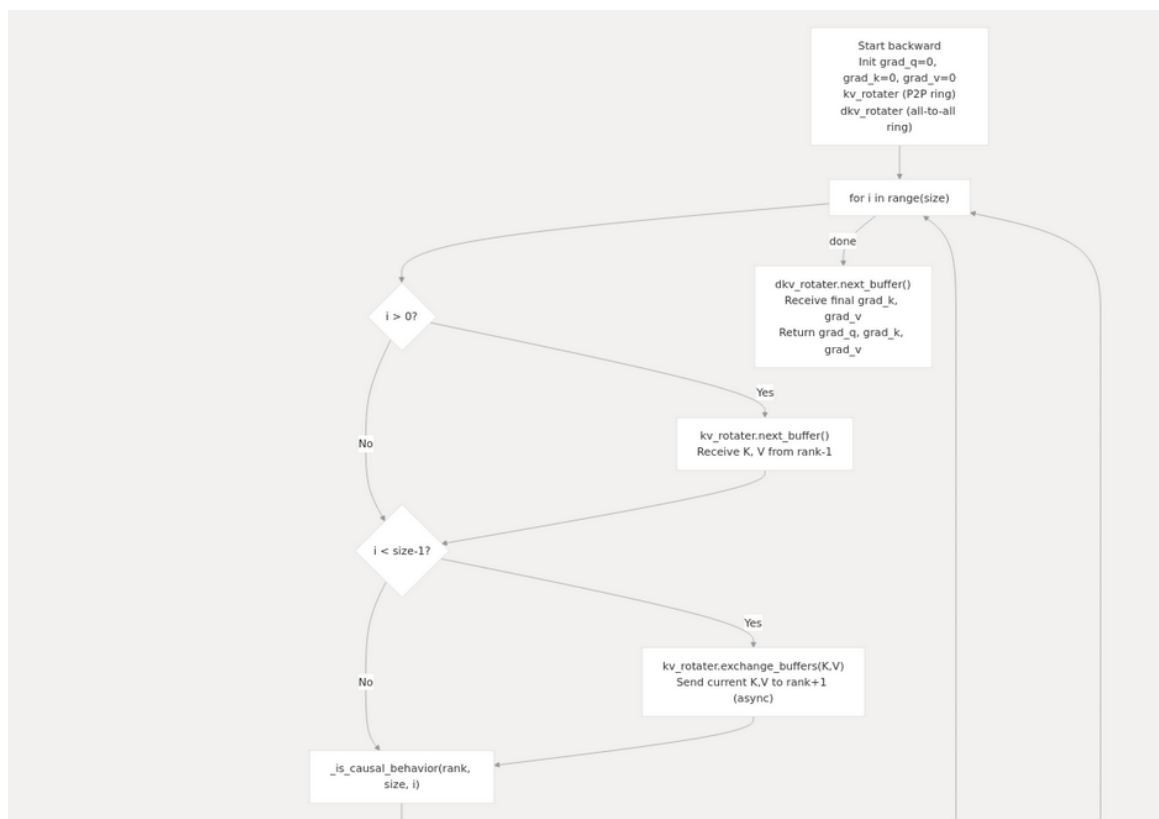
In `_SDPAMerger._merge_one`, `_partial_update` is used when `partial=True` to update only the second chunk of accumulated results:

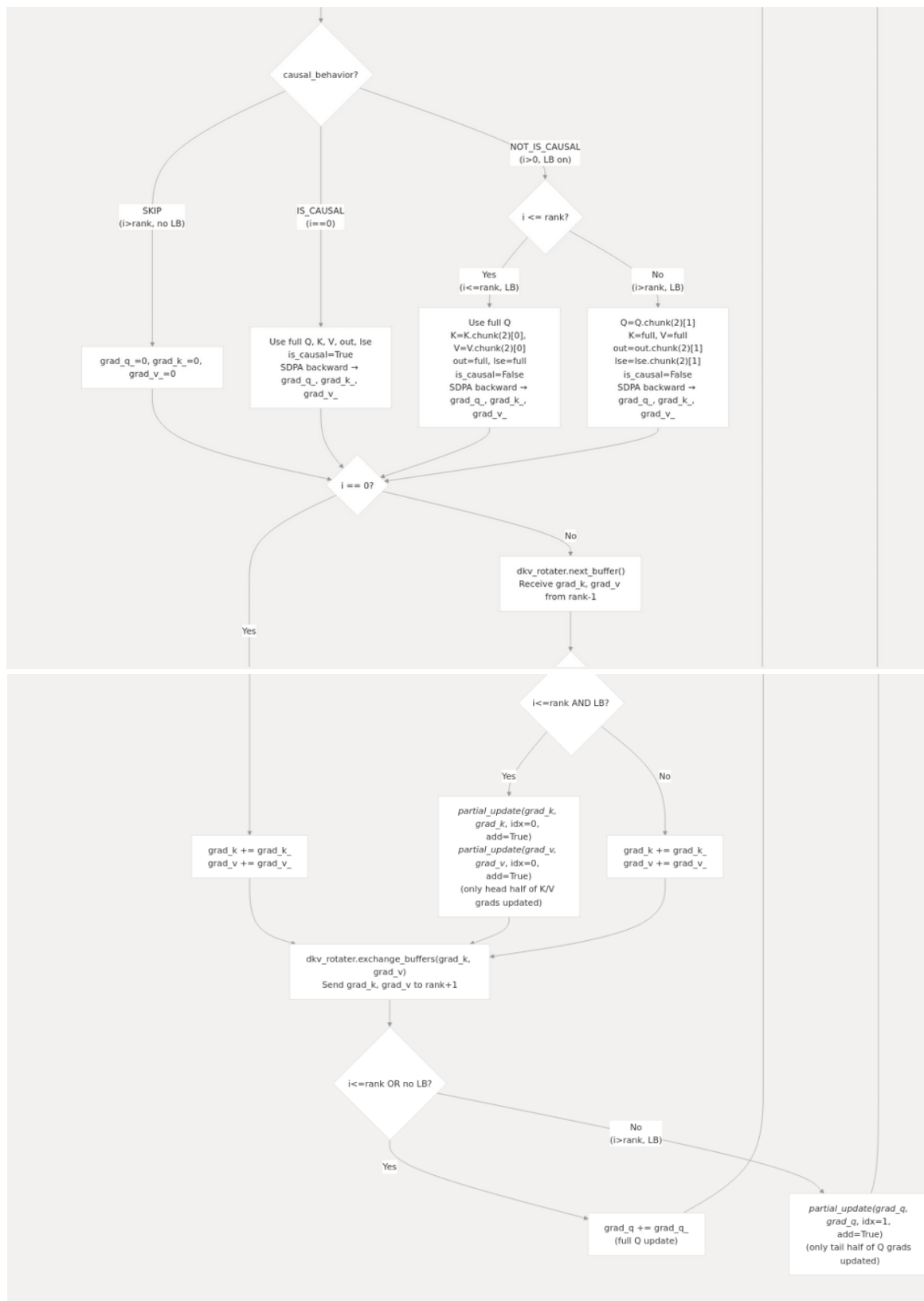
```
if partial:
    self._lse = _partial_update(
        self._lse,
        lse,
        dim=self._seq_dim,
        n_chunks=ROUND_ROBIN_CYCLE,
        idx=1,
        add=False,
    )
    self._out = _partial_update(
        self._out,
        out,
        dim=self._seq_dim,
        n_chunks=ROUND_ROBIN_CYCLE,
        idx=1,
        add=False,
    )
```

Backward Pass :

In the backward pass, `_partial_update` is used the same way but with addition for gradient accumulation with load balancing instead of complete replacement like in forward

```
if i <= rank and _cp_options.enable_load_balance:
    grad_key = _partial_update(
        grad_key,
        grad_key_,
        dim=seq_dim,
        n_chunks=ROUND_ROBIN_CYCLE,
        idx=0,
        add=True,
    )
```

Backward pass for loadbalanced approach with round-robin sub chunking :



Forward pass for loadbalanced approach with round-robin sub chunking :



I am planning to try to implement this into ours with the partial updates this time in backward as well to check if it can improve the throughput while maintaining the loss saturation same as now tomorrow.

Regards

Adhitya Charan